A-A-51581A
31 August 1987
SUPERSEDING
A-A-0051581(DM)
25 July 1986

COMMERCIAL ITEM DESCRIPTION

RESUSCITATOR, HAND OPERATED

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a handpowered, compressible bag-valve-mask resuscitator with an oxygen reservoir suitable for use by medical or paramedical personnel in emergency ventilation of humans.

Salient characteristics:

The resuscitator shall be a bag and mask type, complete with a reservoir, in a carrying case with an instruction manual.

The resuscitator shall consist of, at least, a face mask, a patient valve, a ventilator bag with valving, a pressure limiting system, a nipple for direct attachment to oxygen tubing for operation without oxygen reservoir, and an oxygen reservoir bag.

The materials for the face mask, patient valve housing, compressible ventilator bag, nipple, and oxygen reservoir bag shall be plastics and/or rubber or synthetic rubber as appropriate. Valving elements may be metallic but shall not interfere with the operator's observation of patient gas. Plastic/rubber type materials shall maintain their nominal dimensions, configuration, and operational performance by resisting deterioration caused by aging, adverse storage conditions, and specified temperature extremes. All components shall be in accordance with appropriate ASTM or ANSI Standards. They shall be capable of effective operation at temperatures of -20°C to +50°C and withstanding storage temperatures of -40°C to +70°C. Plastic/rubber type materials for these components shall resist fungal growth and be rated number 0 to number 1 when tested in accordance with ASTM G21, Determinating Resistance of Synthetic Polymeric Materials to Fungi; and shall resist deterioration due to aging in accordance with Accelerated Aging Test specified herein.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:58Z
Check the source to verify that this is the current version before use.

Open-cell foam elastomers are prohibited. The resuscitator, including listed components, shall be capable of being sterilized repeatedly using cold chemical disinfectant liquid(s), and using ethylene oxide or by steam autoclaving at temperatures up to 134°C.

All materials shall be inherently corrosion resistant or treated for corrosion resistance.

The face mask shall be sized to fit a medium size adult face. It shall have a soft rim which seals effectively and conforms well to facial anatomy. The mask shall be transparent clear or have a transparent (clear) center section. It shall have 22/15 mm connectors which shall be in accordance with ASTM Standard F-920, General Requirements.

The patient valve shall prevent rebreathing of the patient's exhaust gas, shall permit spontaneous breathing when ventilating bag is not being operated, shall direct bag pumped gas to the patient, and shall resist jamming despite temperature extremes, presence of debris, moisture or supplemental oxygen inflow as great as 10 LPM. The valve shall be transparent and have a standard 22/15 mm connector, at the face mask end, which shall be in accordance with ASTM Standard F-920. It shall be easy to disassemble/assemble and clean and its components shall be designed/sized to prohibit incorrect assembly. The valve dead space shall not exceed 12 ml.

The ventilator bag shall be single hand operable and capable of performing as specified when operated by a hand which has maximum values listed below:

a. Hand length (wrist joint to tip of longest finger)--6.32 in.
b. Hand breadth (across finger joints in palm)---2.72 in.
c. Hand grip strength (momentary hold, females)--39 lbs.

The ventilator bag shall have a minimum volume of 1000 ml and be capable of delivering 1000 ml per squeeze when operated at a rate of 50 cycles per minute by a hand no greater than listed above. The bag shall be provided with a one-way valve for ambient gas admission.

The pressure limiting system shall be either a single component such as a relief valve or a design feature of combined components such as a double ventilator bag (actual ventilator bag inside a pressure control bag). It shall sense the down-stream pressure (lung pressure) and shall allow as much, but no more than, 70 cm H_2O ($\pm$ 10 cm H_2O) pressure to be delivered. Shall provide an audible or visible warning if activated. Shall provide a means by which the limiting system may be over-ridden. Shall fully comply with ASTM F 920 Standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:58Z
Check the source to verify that this is the current version before use.

The nipple for attachment to oxygen tubing (used to obtain partial oxygen concentration when oxygen reservoir is not being used) shall accept 1/4 inch ID tubing. It shall feed into the resuscitator in a manner to allow partial oxygen concentrations as great as 40 percent when oxygen flow rate is 10 LPM and cycle rate is not less than 12 squeezes per minute at 600 ml per squeeze. This nipple is not required to be accessible when the reservoir is attached to the resuscitator.

The oxygen reservoir bag shall be a separate assembly and shall be capable of being attached to the resuscitator in a manner that 100 percent (-1 percent) oxygen concentration is obtainable when oxygen flowrate is 10 LPM and cycle rate is not less than 12 squeezes per minute at 600 ml per squeeze. The bag shall be airtight and have a maximum capacity of, at least, 1200 ml.

The resuscitator with oxygen reservoir attached shall have a valving design that shall allow ambient air to enter the ventilator bag if insuffient oxygen is available to fill the ventilator bag when ventilator bag is empty or is expanding. The design shall allow surplus oxygen to escape into the atmosphere when the ventilator bag is full or being squeezed.

The instruction manual shall describe the theory of operation and inlcude operating instructions including procedures to obtain various oxygen concentrations; assembly, disassembly, and sterilizing instructions; a list of replacement parts by manufacturer's catalog numbers; descriptive sketches as appropriate; statement of warranty; and manufacturer's and distributor's name, address, and phone number.

The carrying case shall be semi-rigid plastic construction and include a handle. It shall house and protect the complete resuscitator and instruction manual during transport and storage. When the carrying case is being held by its handle and contains the complete resuscitator, it shall be capable of withstanding a drop (handle released) from 3 feet onto a concrete floor without opening and not suffering damage to the case or content (minor scuff mark acceptable). The case, while in up-right position, closed, containing complete resuscitator, and resting on a concrete floor, shall be capable of withstanding a top-load of 200 pounds (step-on test) without opening or suffering damage to the case or contents. The cubic foot of the case shall not exceed 0.35 cubic foot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:58Z
Check the source to verify that this is the current version before use.

In addition to being in strict conformance with this specification, the resuscitator shall comply with ASTM F 920 Standard for Minimum Performance and Safety Requirements for Resusicators Intended for Use with Humans. In the event of conflict, this specification shall prevail.

Workmanship. The resuscitator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One resuscitator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Test procedures shall be in accordance with ASTM F 920, and as specified herein. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:58Z
Check the source to verify that this is the current version before use.

Examination. Examination shall be conducted to determine compliance with specification requirements and shall include, but not be limited to, presence of all components, face mask and rebreathing valve transparency, size and softness of face mask, content of instruction manual, and general appearance and workmanship.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105 and as specified herein. The unit of product for inspection purposes shall be one resuscitator. Except for fungal test, ASTM G21, and Accelerated Aging Test specified herein, inspection level shall be S-2 for both examination and testing, and AQL (defects per 100 units) for examination shall be 1.5.

AQL for testing shall be No. 1. If identical units have passed the specified fungal test within the past 3 years, then a certification of fungistatic compliance, accompanied with the test report, shall be acceptable in lieu of this test. Otherwise, one randomly selected unit per contract quantity shall be tested for fungistatic compliance as specified. Five randomly selected unit per contract quantity shall be tested in accordance with Accelerated Aging Test specified herein. If identical units have passed the specified aging test within the past 3 years, then a certification of aging compliance, accompanied with test report, shall be acceptable in lieu of this test.

Accelerated Aging Test. All plastic, synthetic rubber, and rubber components which are flexible shall be tested for aging characteristics following test procedures described in ASTM D 573, Standard Test Method for Rubber-Deterioration in an Air Oven. Test specimens shall be obtained by severing each component to be tested, forming a pair of twin specimens, then marking the pairs for identification purpose. One twin of each pair shall be selectable as an aging specimen and the other selectable as the corresponding unaged/original specimen. Comparative values of properties shall be restricted to comparing twins within a pair of specimens. Properties to be compared shall be ultimate elongation and tensile strength. A comparative reduction in excess of 25 percent of either property shall be a failure.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:58Z
Check the source to verify that this is the current version before use.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6515-01-061-7812

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:58Z
Check the source to verify that this is the current version before use.

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4093

Location: CID/RESUS/H12-12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:58Z
Check the source to verify that this is the current version before use.